ETRO ELECTRONICS & INFORMATICS

Brussels, November 30, 2023

Analog Electronics Exercises

Session 4

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Exercise 1

a) Resistors R_1 and R_2 connected in series are at equilibrium temperature T. P_{12} is the noise power transferred from resistor 1 to 2 and vice versa is P_{21}. Calculate P_{12} and P_{21} for an arbitrary bandwidth B. What is the relationship between P_{21} and P_{12}?

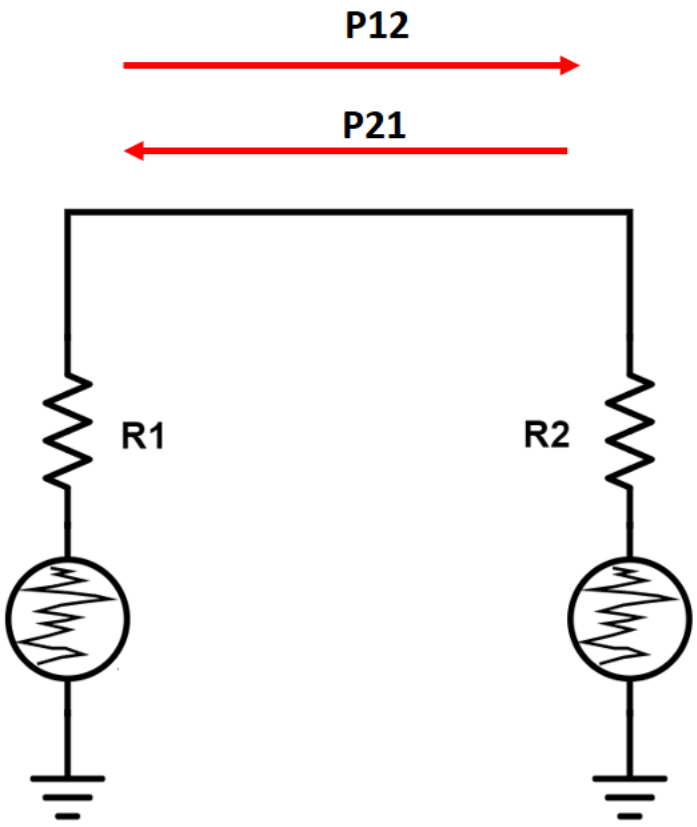


Figure 1: Resistor R_1 and R_2 connected in series

b) In Figure A, calculate integrated noise across the capacitor. Also calculate the same when resistor R is increased to $2R$ as in Figure B? (Hint: integrated noise $\overline{v_n^2} = \int_B \overline{dv_n^2} df$, where B is the bandwidth of interest, which in this case is from 0 to ∞.)

c) Plot the noise power spectral density in both cases across the capacitors with frequency? Take $R = 1K$ and $C = 1pF$, $T = 300K$, $k = 1.38{*}10^{-23}$. In the second case change the value of the resistance to 2K.

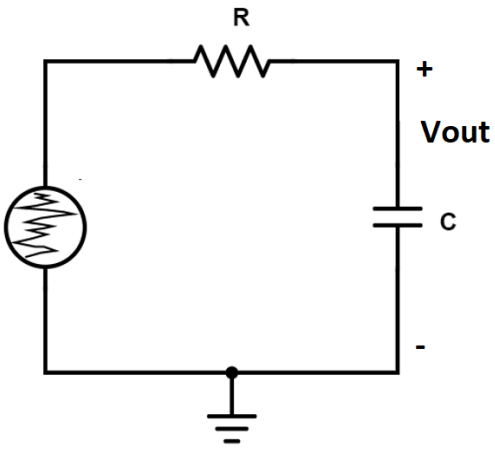


Figure A

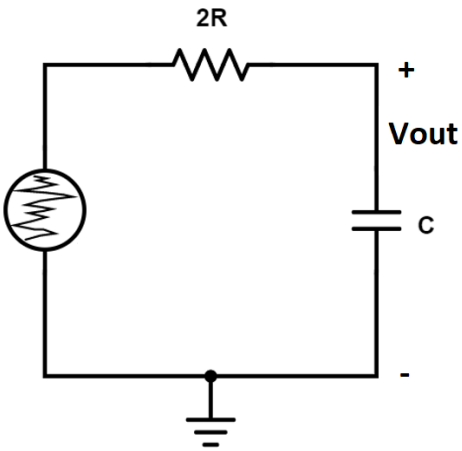


Figure B

Figure 2: RC circuit

Exercise 2

a) Calculate output and input-referred noise of common source amplifier. Assume a transistor with infinite output impedance and a transconductance g_m. Consider only thermal noise for calculations.

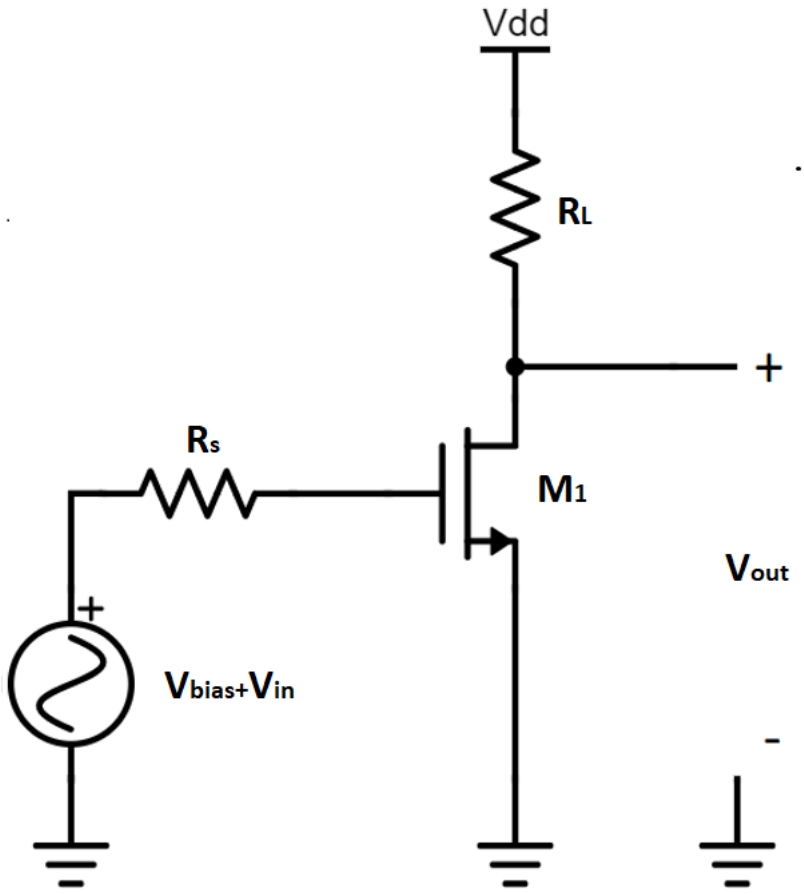


Figure 3: Common source Amplifier

b) Calculate output and input-referred noise of the degenerated amplifier in Fig. 4a. Assume a transistor with infinite output impedance and transconductance g_m. Is the input-referred noise higher or lower compared common source amplifier having similar g_m? Consider only thermal noise for calculations.

c) Calculate output and input-referred noise of the common drain amplifier in Fig. 4b. Assume a transistor with infinite output impedance and transconductance g_m. Consider only thermal noise for calculations.

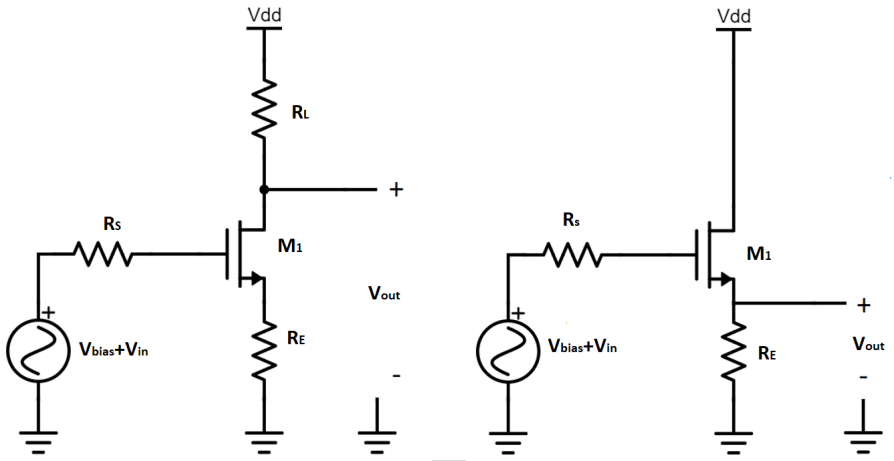


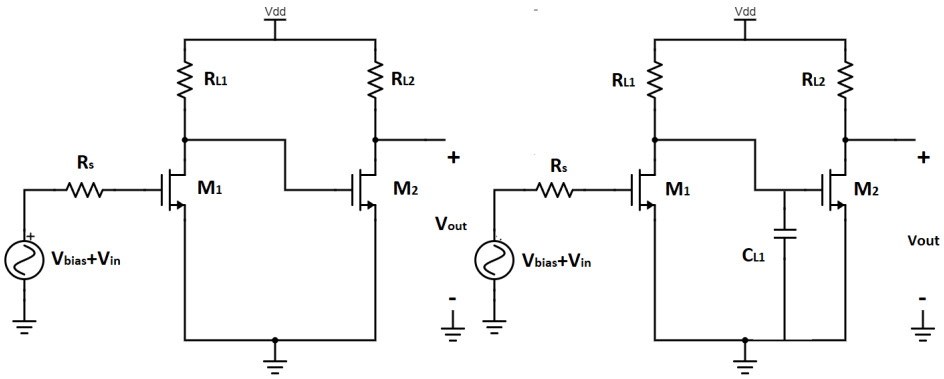
Figure 4: (a) Amplifier with degeneration resistance R_E. (b) Common drain amplifier.

Exercise 3

a) Calculate the input referred noise of the cascaded common source amplifier as shown in Fig. 5a. The transistors M_1 and M_2 have transconductance g_{m1} and g_{m2}. Assume only thermal noise and neglect the flicker noise of transistors. Also assume infinite output impedance with all transistors in saturation.

b) What are the noise sources contributing most of the noise? From the expression do you see that in order to decrease the noise contribution of the second stage you can increase the gain of the first stage? Is there any limit to reduce the noise contribution from the second stage (Hint: Take into account channel length modulation and hence finite output impedance/intrinsic gain of the transistor)? Any technique you can try to overcome the limit?

c) Now add capacitance C_{L1} to the output of the first stage as shown in Fig. 5b. How will it change the input referred noise contribution of the second stage with frequency?



(a) (b)

Figure 5: (a) Cascaded common source amplifier. (b) Cascaded amplifier with capacitive load in first stage.

Exercise 4

Fig. 6 shows a common source amplifier with gate to source capacitance C_{gs}. In the figure, $R_L = 5K$, $R_S = 100K$, $C_{gs} = 33f$ and $g_m = 1mS$. For transistor M1 drain flicker noise current spectral density at 1Hz=$5 * 10^{-15} A^2/Hz$.

a) Plot the output and input noise voltage spectral density with frequency (by hand or MATLAB). In this question take into account both thermal and flicker noise contribution. Take $\gamma = \frac{2}{3}$.

b) The input signal bandwidth is from 10MHz to 100MHz. Calculate the input referred integrated noise in the given bandwidth and also minimum detectable signal value.

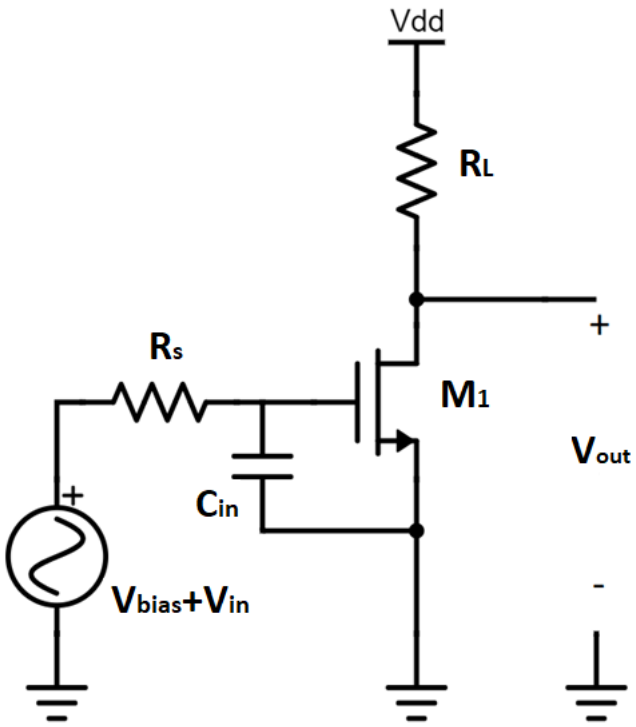


Figure 6: Common source Amplifier

Exercise 5

Fig. 7 shows a different pair. Transistor M_1 and M_2 have transconductances g_m.

a) Assume bias current from transistor M_3 does not add any noise. What is total input referred noise of the differential pair? Take into account only thermal noise.

b) Now imagine transistor M_3 adds noise. Do you think it will add noise to the differential output? Can flicker noise in M_3 change the gain of the amplifier with time? Assume that the input is very small signal.

c) What do you think will be the impact of noise in supply Vdd?

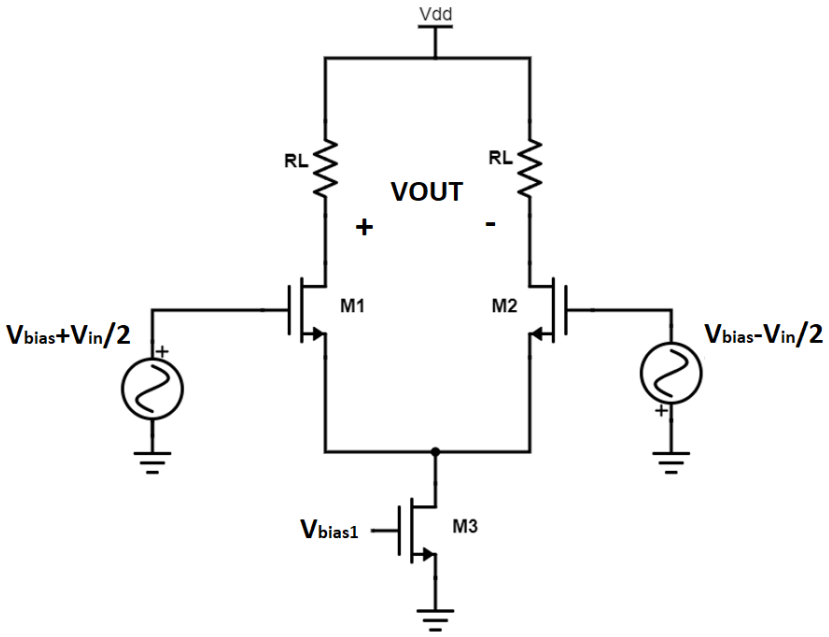


Figure 7: Differential Amplifier

Exercise 6

Fig. 8 shows a 2-stage OTA. Assume each of the transistors are in saturation and take into account the effect of output impedance and thermal noise of the transistors.

a) Calculate the input referred noise contribution at lower frequency from the transistors M_{1A} , M_{1B} , M_{2A} and M_{2B}. Assume the transconductance of M_{1A} and M_{1B} is g_{mn} and transconductance of M_{2A} and M_{2B} is g_{mp}.

b) Can you comment on noise contribution of transistors M_3 and M_4 at very low frequencies. At what frequency approximately will the 2nd stage noise will begin to dominate (Hint: Frequency at which first stage gain drops to 1). Take low frequency output impedance of first stage as R_{o1} and R_{o2}.

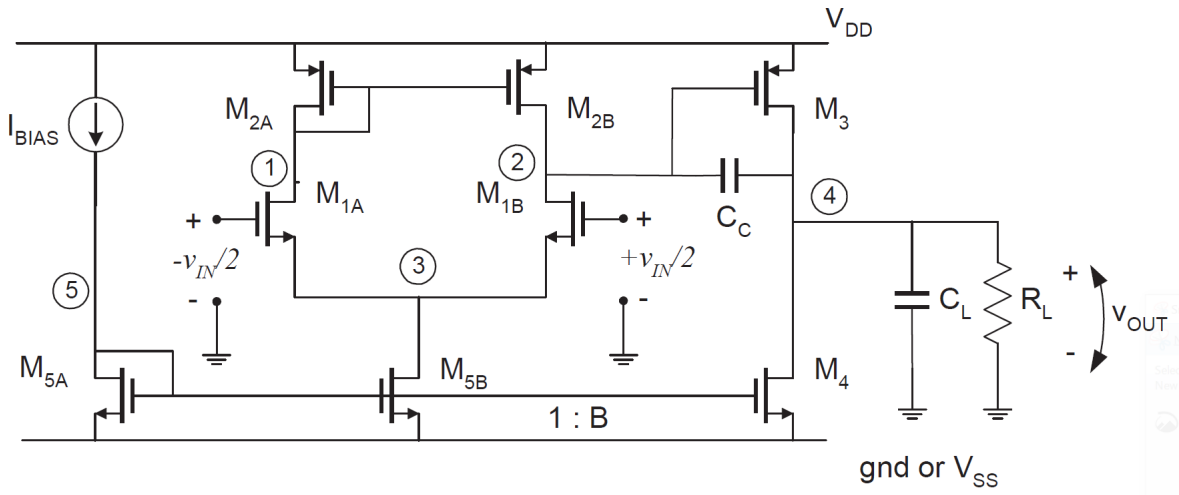


Figure 8: 2 Stage OTA

MATLAB session

`session4_sol1.m`: Fix V_{OV}, increment I_{BIAS} at each step, plot how the current, width, thermal noise, and flicker noise change with g_m.
`session4_sol2.m`: Find the corner frequency, at which the thermal noise and flicker noise spectral densities are equal.
`session4_sol3.m`: Fix I_{BIAS}, increment V_{OV} at each step, plot how the width, thermal noise, and flicker noise change with g_m.